UNIT 1 · TOPIC 1(A)

Status of Farm Mechanization in India

Syllabus topic: Status of farm mechanization in India; power availability on farms and trends in mechanization

What farm mechanization means

Farm mechanization means using engineering and technology inputs — mechanical, electrical and renewable power sources with matching implements — to carry out farm operations on time and efficiently. It cuts drudgery and cost, and raises productivity.

Benefits

- Timely operations — saves **15–20%** of inputs such as seed and fertilizer.
- Less drudgery for farm workers.
- Precise metering and placement of inputs.
- Higher cropping intensity.
- Lower cost of operation per unit of output.
- Makes conservation agriculture possible.

How mechanized is India?

India's overall level of farm mechanization is **~45–47%**, compared with **~95%** in the USA, **~75%** in Brazil and **~57–60%** in China.

Operation	Mechanization level (approximate)
Soil working & seedbed preparation	~40%
Seeding & planting	~29–30%
Plant protection	~34–35%
Irrigation	~37%
Harvesting & threshing (wheat, rice)	60–70%
Harvesting (other crops)	<15–20%

The least mechanized operations remain **rice transplanting, cotton picking, sugarcane harvesting** and horticultural operations.

Tractor industry

- India is the world's largest tractor manufacturer and market. Domestic sales crossed **~1.0 million units in 2025** — about one-third of global production.
- Major manufacturers: Mahindra & Mahindra (largest, ~40%+ share), TAFE, Sonalika (ITL), Escorts Kubota, John Deere India, New Holland.
- The best-selling power bands are **31–40 hp** and **41–50 hp**.
- Tractor density is highest in the north-western states — Punjab, Haryana and western UP.

Government programmes

- **SMAM** (Sub-Mission on Agricultural Mechanization) — launched **2014–15**. Components: Custom Hiring Centres (CHCs), Hi-Tech Hubs, Farm Machinery Banks, subsidies of 40–50% (up to 80% for CHC/group ownership), training and testing.
- **CRM Scheme** (Promotion of Agricultural Mechanization for In-Situ Management of Crop Residue) — **2018**. Promotes the Happy Seeder, Super SMS, mulcher, zero-till drill and balers to curb residue burning in Punjab, Haryana, UP and NCT Delhi.
- **FARMS mobile app** (Farm Machinery Solutions) — **2019**; brings "Uberisation" to custom hiring.

Institutions

India runs four Farm Machinery Training & Testing Institutes (FMTTIs):

Institute	Location	Established
Central FMTTI — tractor testing (national tractor test code)	Budni (Madhya Pradesh)	1955
Northern Region FMTTI	Hisar (Haryana)	1963
Southern Region FMTTI	Garladinne, Anantapur (Andhra Pradesh)	1983
North-Eastern Region FMTTI	Biswanath Chariali (Assam)	1990

- **CIAE Bhopal** (ICAR-Central Institute of Agricultural Engineering, est. **1976**) does R&D on farm equipment and runs the AICRP on Farm Implements and Machinery (FIM), the AICRP on Ergonomics & Safety (ESA) and the AICRP on Utilization of Animal Energy.

Safety law

The **Dangerous Machines (Regulation) Act, 1983** covers power threshers, chaff cutters and sugarcane crushers to protect operators.

Practice MCQs

Q4. Overall level of farm mechanization in India is about:

(a) 25–30%
(b) 45–47%
(c) 60–65%
(d) 75–80%

Q5. SMAM was launched in:

(a) 2010–11
(b) 2014–15
(c) 2016–17
(d) 2018–19

Q6. The Central Farm Machinery Training and Testing Institute is located at:

(a) Hisar
(b) Budni
(c) Anantapur
(d) Bhopal

Q7. Power threshers and chaff cutters are regulated for operator safety under:

(a) Factories Act 1948
(b) Dangerous Machines (Regulation) Act 1983
(c) Insecticides Act 1968
(d) BIS Act 1986

Q8. Which operation has the LOWEST mechanization level in India?

(a) Seedbed preparation
(b) Wheat harvesting
(c) Rice transplanting & cotton picking
(d) Irrigation

Answers

Q4 — (b) 45–47%

Q5 — (b) 2014–15

Q6 — (b) Budni

Q7 — (b) Dangerous Machines (Regulation) Act 1983

Q8 — (c) Rice transplanting & cotton picking